

這份是 HiABC 計畫的住院急性惡化子研究 **HiABC-exacerbation Proposal** 精要版，與 52 週主試驗 HiABC-stable 並行設計，供團隊討論與院內 proposal synopsis 使用。

HiABC-exacerbation Trial

Hypertonic Saline and Active Cycle of Breathing Technique during Hospitalized

Bronchiectasis Exacerbation:

A Randomized Controlled Pilot Trial

1. Background and rationale

During a pulmonary exacerbation, patients with bronchiectasis have increased sputum volume and viscosity, worse cough and more difficulty expectorating, and they are often hospitalized for intravenous antibiotics. Airway clearance is routinely advised during admission, but which inpatient airway-clearance strategy works best has not been established.

The companion 52-week trial, **HiABC-stable**, tests whether nebulized 3% hypertonic saline combined with ACBT prolongs the time to first exacerbation in **clinically stable** patients. It does not address whether the same strategy is feasible, tolerable and helpful during the acute phase, when bronchospasm risk is higher and sputum burden is greatest.

HiABC-exacerbation is an inpatient pilot trial covering admissions of ≥ 7 days. It will generate efficacy, safety and feasibility data for the HiABC strategy during exacerbation, and it provides a structured pathway for patients to be referred into HiABC-stable after recovery.

2. Study hypothesis

In adults hospitalized for a bronchiectasis exacerbation, **3% hypertonic saline and standardized ACBT in a single daily session** added to usual inpatient care will increase sputum clearance (sputum wet weight) and improve cough-related quality of life, and may shorten length of stay, compared with **0.9% Isotonic Saline + Pragmatic Usual BE Care**.

3. Study design

Prospective, randomized, controlled, parallel-group pilot trial with blinded sputum-weight measurement. Single- or dual-center.

Randomization: 1:1, within 48 hours of admission

Control: 0.9% Isotonic Saline + Pragmatic Usual BE Care	HiABC intervention
0.9% isotonic saline 5 mL QD	3% hypertonic saline 5 mL + standardized ACBT, QD (sequence individualized)

Every day (no rest day), from Day 1 until discharge (admission ≥ 7 days)	Every day (no rest day), from Day 1 until discharge (admission ≥ 7 days)
Pragmatic usual inpatient BE care	Usual inpatient BE care

Control arm description

- Nebulized 0.9% isotonic saline 5 mL once daily, every day during admission, using the same nebulizer and schedule as the intervention arm.
- **Pragmatic usual inpatient BE care:** antibiotics, bronchodilators, oxygen and all other management are left to the treating team according to routine practice. Participants may continue their usual self-directed airway clearance, and ward-routine chest physiotherapy is allowed and recorded.
- The control arm does **not** receive structured ACBT training or supervised ACBT sessions from the study team.

Safety run-in for 3% HS (both arms, blinded)

- A short-acting bronchodilator is given before every nebulization.
- The first dose is supervised, with SpO₂ and symptoms monitored and FEV₁ or peak flow measured before and after the dose when feasible.
- If there is a fall of $\geq 15\%$ in FEV₁/PEF, a significant drop in SpO₂, or clinical bronchospasm, study nebulization is stopped. The patient stays in the analysis (intention-to-treat).

Intervention period: from randomization **until discharge**; the admission is expected to last **≥ 7 days**. Treatment is given **every day during admission, with no rest day**. The study intervention **stops at discharge**.

4. Study population

Major inclusion criteria

- Age ≥ 18 years
- CT-confirmed bronchiectasis
- Hospitalized for a pulmonary exacerbation meeting the prespecified consensus definition (worsening respiratory symptoms for ≥ 48 hours requiring antibiotics)
- Productive of sputum at enrollment
- Randomized within 48 hours of admission
- Expected hospital stay ≥ 7 days (e.g. planned ≥ 7 days of intravenous antibiotics)
- Able to perform ACBT with coaching

Major exclusion criteria

- Cystic fibrosis
- ICU admission, or need for invasive or non-invasive ventilation

- Hypercapnic respiratory failure, or hemodynamic instability
 - Major/recent hemoptysis
 - Pneumothorax
 - Unstable asthma / significant bronchospasm, or a failed 3% HS safety run-in
 - Current regular hypertonic saline use
 - Current regular structured ACBT or equivalent intensive ACT
 - Admission primarily for another diagnosis (e.g. pneumonia with new consolidation, heart failure, pulmonary embolism)
 - Active cavitary NTM disease with recent major hemoptysis, or NTM multidrug therapy newly started or about to be substantially changed during admission
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5. Primary endpoint

Mean 24-hour sputum wet weight (g) from Day 7 to Day 14

- Only sputum collected **during the hospital stay** is used. For participants discharged before Day 14, the window runs from Day 7 to the day of discharge; no sputum is collected after discharge.
- Sputum is collected into pre-weighed containers and weighed by staff who do not know the patient's allocation.
- The analysis compares groups using a mixed model for repeated measures (MMRM) of daily 24-hour sputum wet weight from Day 7 to Day 14, adjusted for baseline (Day 0) 24-hour sputum wet weight. The model uses all available in-hospital days, so participants discharged before Day 14 remain in the analysis; a sensitivity analysis will examine the effect of early discharge.

Rationale: sputum wet weight is an objective, short-term marker of airway clearance and is responsive during admission. The Day 7–14 window measures the sustained effect of the regimen after the first week of treatment, rather than the early acute phase. LCQ and length of stay are carried as key secondary outcomes rather than co-primary, to avoid multiplicity in a small pilot.

6. Key secondary outcomes

- Change in LCQ total score from admission to Day 7 and to discharge
- Hospital length of stay
- Mean 24-hour sputum wet weight from Day 1 to Day 6 (early phase)
- Sputum wet weight per treatment session (during the session and 30 minutes after it)
- Ease of expectoration (VAS) and sputum color
- Time to clinical stability
- CRP / WBC trajectory
- 30-day readmission
- Bronchospasm, desaturation, hemoptysis and other adverse events
- Treatment adherence and completed sessions
- Feasibility: screening-to-randomization rate, retention, completeness of sputum collection

- Proportion referred to and enrolled in HiABC-stable

7. HiABC intervention

Participants in the HiABC-exacerbation intervention arm will receive a structured, daily airway-clearance regimen consisting of nebulized 3% hypertonic saline (5 mL) and standardized ACBT (approximately 10–20 minutes) every day during admission, with no rest day.

The sequence of nebulization and ACBT during the clearance session may be individualized according to participant comfort, baseline mucus burden, and clinical tolerance. Standardized instruction will ensure that both components are completed within the designated daily session.

Standardized ACBT cycle:

breathing control → thoracic expansion → breathing control → forced expiration/huff → repeat as required.

ACBT is delivered by the same trained respiratory therapists/physiotherapists and instructional video as in HiABC-stable. Sessions are adapted to the patient's dyspnea and fatigue during the acute phase and each daily session is supervised.

8. Sample size

N = 50–60 participants (25–30 per arm)

As a pilot trial, the sample size is chosen mainly for feasibility and to estimate effect size and variance for a future definitive study. For reference, 26 participants per arm give 80% power (two-sided $\alpha = 0.05$) to detect a standardized difference of about 0.8 in the primary endpoint. With about 10–15% attrition, this corresponds to 56–60 participants.

A standardized difference of 0.8 is a large effect. The final target will be confirmed using local data on the mean and SD of inpatient 24-hour sputum wet weight.

9. Schedule and referral to HiABC-stable

Admission → Screening + safety run-in → Randomization (≤ 48 h) → Day 1 → Day 7 → until discharge (≥ 7 days) → Discharge → Week 4 post-discharge visit

- **Daily:** 24-hour sputum wet weight, adverse events and adherence
- **Admission, Day 7 and discharge:** LCQ, ease of expectoration, CRP/WBC and spirometry (when feasible)
- **30 days:** readmission status

Referral pathway: at the Week 4 post-discharge visit, participants who have been **clinically stable for ≥ 4 weeks since discharge** are assessed against the HiABC-stable eligibility criteria and, if eligible and willing, referred for enrollment into the 52-week main trial. HiABC-stable randomization is independent of the HiABC-exacerbation allocation, and prior participation is recorded and balanced at randomization.

10. Comparison of the two HiABC populations

	HiABC-stable	HiABC-exacerbation
Population	Clinically stable ≥ 4 weeks; ≥ 2 exacerbations in the past 12 months	Hospitalized for an acute pulmonary exacerbation
Setting	Outpatient / home, multicenter	Inpatient ward, single/dual-center
Design	Definitive RCT	Pilot RCT
Sample size	$N \approx 240$ (240–300)	$N \approx 50$ –60
Intervention	3% HS 5 mL + ACBT in one daily session, 6 days/week; sequence individualized	3% HS 5 mL + ACBT in one daily session, every day during admission; sequence individualized
Control	0.9% Isotonic Saline + Pragmatic Usual BE Care (same in both studies)	
Duration	52 weeks	Admission ≥ 7 days (intervention to discharge) + 4-week post-discharge visit
Primary endpoint	Time to first pulmonary exacerbation	Mean 24-h sputum wet weight, Day 7–14 (in-hospital)
Key secondary	Annualized exacerbation rate, QoL-B, LCQ	LCQ change, length of stay, safety
Question answered	Does HiABC prevent exacerbations?	Is HiABC safe and does it improve clearance during an exacerbation?

11. Proposed primary objective

To determine whether nebulized 3% hypertonic saline combined with standardized ACBT increases sputum clearance, measured as mean 24-hour sputum wet weight from Day 7 to Day 14 of the hospital stay, compared with nebulized 0.9% Isotonic Saline + Pragmatic Usual BE Care in adults hospitalized for a bronchiectasis exacerbation.

12. Proposed key message

HiABC-exacerbation extends the HiABC airway-clearance strategy to the acute inpatient setting. In a safety-monitored inpatient pilot RCT (admission ≥ 7 days), it will test whether combining mucus hydration with active secretion clearance improves sputum clearance and cough during hospitalization. It also serves as a structured entry pathway into the 52-week HiABC-stable trial.